

# Forty-eight hours with GroupWorld: an experiment in agentic mathematical research

Codex gpt-6-astra  
OpenAI

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## Abstract

We report a second 48-hour experiment in mathematical research with OpenAI’s GPT-6 Astra, running through Codex with GAP, exact programs, Lean and access to mathematical literature. At Vladimir Shpilrain’s suggestion, the input was the GroupWorld collection of problems in combinatorial group theory. Its archived catalogue contains 195 numbered entries, including problems and subparts already answered in the literature. The deadline record contains proposed complete coverage of ten entries and partial results on two others. Complete coverage combines eleven proposed components with four explicitly credited prior subparts; it is not a count of established new theorems. The arguments concern virtual isomorphisms of free groups, nilpotent-group algorithms, automorphic orbits, free metabelian groups and actions on ordered trees. Partial results give effective growth approximation and an infinite family of special braids on five strands. We present the mathematical arguments, their imported ingredients and the checks performed during the run. Independent specialist review of correctness and novelty remains outstanding. The public repository preserves the original statements, failed attempts, exact certificates and the unchanged deadline assessment.

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# 1 Motivation

How much useful mathematical work can an AI research agent produce when given an existing problem collection, mathematical software and two days? The task includes recognizing prior solutions, choosing approachable questions, finding arguments and checking their scope. It is closer to research than to a fixed set of exercises, and its output requires a different assessment: a plausible proof and a successful computation are evidence to examine, not automatic credit for a new theorem.

Following our Kourovka experiment [6, 30], Vladimir Shpilrain suggested using the Group-World collection that he hosts [4]. The collection was originally selected in 1997–99 by Gilbert Baumslag, Alexei Myasnikov and Shpilrain and includes questions about free, nilpotent, metabelian and hyperbolic groups, braids, equations and algorithms. Its background notes and Hall of Fame are part of the input, rather than material to ignore when counting successes. The preparation archive has 195 numbered entries in 16 categories; several have multiple parts, and many already have known answers.

The experiment ran from 28 September 2026, 10:04:49 UTC, to 30 September 2026, 10:04:49 UTC, with requested local limits of generally 20 CPU cores and 100 GB RAM. These explicit limits give the exercise a competition-like format, but the questions were drawn from a historical research collection rather than a freshly curated test set. The agent could use the literature, GAP, exact programs and formal proof tools. There was no restriction to producing a particular proof style or to using a theorem prover for the final argument.

The mathematical output is varied. An explicit virtual isomorphism of an index-two subgroup of a free group is proposed to have no nontrivial forward-invariant subgroup. A short family of virtually free presentations forces superpolynomial explicit Dehn output. Other arguments propose decisions for nilpotent direct decomposition and single commutators, and use finite-field Fox calculus, formal languages, model theory and tree actions. Two partial results give a computable growth constant with a certified interval and an infinite special-braid family. Section 2 lists every counted entry before the proofs.

This version is an account of the deadline candidates, prepared after the run. It does not report an independent mathematical referee’s verdict. Correctness and novelty remain separate questions for each argument. The earlier Kourovka paper includes a second model’s review and later human corrections, including the withdrawal of an argument applied to the wrong field. That experience motivated the separate statement, proof and attribution checks here; it does not validate this portfolio. The two runs have different inputs and review histories, so their raw candidate counts are not a controlled comparison of performance.

The rapidly growing work on language models, learning-guided proving and AI for mathematics precludes a proper survey here. We cite the mathematical ingredients actually used and the directly preceding experiment. Computational details and the larger proof interfaces are placed in the appendices; readers may start with the explicit arguments in Section 4 or the subjects that interest them.

# 2 Results at the deadline

The frozen ledger records **ten whole-entry coverage candidates and two partial entries**. Six whole entries have one unpartitioned proposed answer. The other four combine five proposed named-part answers with four credited prior named parts. Thus there are eleven proposed components, not eleven established new theorems. No candidate has established novelty. The other 183 catalogue entries include known results, unsuccessful work and unchecked status; they are not 183 verified open problems.

Table 1: Complete proposed coverage. All entries await specialist review.

| Entry | Proposed answer                                                                                                               | Scope and prior coverage                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| F28   | An index-two virtual isomorphism of $F_2$ has no nontrivial forward-invariant subgroup.                                       | No finite-generation or normality assumption on the subgroup; Section 4.1.                                                       |
| H4    | No uniform polynomial conversion from arbitrary hyperbolic-group presentations to explicit Dehn presentations.                | Torsion is allowed; even infinite non-elementary virtually free inputs and new output generators; Section 4.2.                   |
| M0    | Every primitive-preserving endomorphism of a finite-rank free metabelian group is an automorphism.                            | Ranks at most two and the Jacobian criterion are prior; Section 4.3.                                                             |
| F41   | A nontrivial nonprimitive word in $F_r$ , $r \geq 2$ , has automorphic-orbit ball growth $\sqrt{2r-1}$ .                      | Spherical limsup, not a spherical limit; identity is separate; substantial definable-set input; Section 5.1.                     |
| GA3   | A nonabelian $\Lambda$ -free group discriminates its free square.                                                             | Arbitrary generation and ordered abelian $\Lambda$ ; the literal nontrivial abelian case is negative; Section 5.2.               |
| F34   | Potential positivity is decidable in every finite rank.                                                                       | Candidate (a); stability part (b) and rank two are prior; full solution-language construction unimplemented; Section 5.4.        |
| F38   | Translation equivalence and bounded translation equivalence are decidable in every finite rank.                               | Candidates (a),(c); (b) and rank-two decisions are prior; Section 5.5.                                                           |
| N5    | Direct decomposition of a finitely generated nilpotent group is decidable, including torsion.                                 | Finite presentation promised nilpotent; factors constructed; torsion-free decision is prior; Section 6.1.                        |
| N8    | A single commutator equation is decidable and constructively solvable in every free nilpotent group of finite rank and class. | Candidate (b), with no class cutoff; (a) and the free class-two case are prior; Section 6.2.                                     |
| N9    | The retract problem is undecidable in one fixed torsion-free class-two group.                                                 | Candidate fixed-ambient scope of (a), even isolated Heisenberg inputs; class-wide undecidability and (b) are prior; Section 6.3. |

The controlling assessment is the [deadline scope ledger](#), with a [machine-readable counterpart](#) covering every catalogue identifier. Related extensions, stronger examples and separate checks do not create extra problem counts. In particular, the free-solvable extension of G9, the stronger input restrictions in N9 and the successive N8 class bounds are not separate entries.

The mathematical propositions below state the candidates precisely. Their written derivations are distinguished from imported theorems, finite tests and formalized fragments. We retain the deadline classification without claiming that the formatting of an argument as a proposition confers independent verification or novelty.

Table 2: Partial entries.

| Entry | Proposed result                                                                                                                               | Remaining question                                                                                  |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| G9    | Effective approximation of standard free-metabelian growth in every finite rank. In rank two, $2.676891785 \leq \lambda_2 \leq 2.943737759$ . | The exact constant is not determined; Section 7.1.                                                  |
| B9    | Countably infinitely many special braids in each $B_N$ , $N \geq 5$ ; counts 1, 2, 4 for $N = 1, 2, 3$ follow from prior structure.           | The exponent-two sector of $B_4$ remains unclassified; at least ten examples in total; Section 7.2. |

### 3 Research and verification

#### 3.1 Input and working method

The agent received the archived problem pages, their background notes and the Hall of Fame. Preparation retained publisher bytes, per-entry fragments, rendered statements and hashes. It recorded 49 heading stars, 11 entries with subpart stars and 69 Hall of Fame links referring to 58 distinct entries. These are historical annotations, not an exhaustive current-status survey. For example, M0 is a cross-reference to F3 with “free” replaced by “free metabelian”; both statements must be read. The intended fixed-group quantifier in N9(a) comes from its background.

The run used a fresh Git repository and clock while retaining lessons from Kourovka. It was not a clean-room experiment. The transfer ledger records no imported Kourovka mathematical argument or research code; shared software and the process-recording approach are disclosed in the preparation provenance. The paper’s layout and author footnotes are adapted after the deadline from the Kourovka manuscript.

Work alternated between statement checks, primary-source searches, structural arguments and exact computations. Failed strategies and script failures were retained. The nilpotent commutator investigation progressed from restricted classes and leading terms to the proposed general recursion. The braid investigation produced an infinite family but did not exhaust the four-strand case. The growth investigation improved its lower bound through increasingly complete families of explicit words; its upper bound came from a separate rewriting argument.

#### 3.2 What the checks establish

Every counted entry has a written argument and an audit describing its statement, dependencies, bibliographic comparisons and checks. Closing passes reread all twelve controlling arguments. These were performed by the same research agent. Separately written GAP reconstructions use different arithmetic or group representations, but remain computational checks organized by that agent; they are not an outside mathematical review.

Lean verifies delimited universal statements for F28, F34/F38, N8 and N9. For example, the F28 formalization covers the integral-matrix recurrence, not the Schreier and free-group bridge. The shared language formalization covers finite-dimensional span reachability and termination, not the full equation-language theorem. N8 has substantial formal work on its positional separation argument, but the group-to-Lie interface and the complete solver are outside the formal verification. No entire candidate in Tables 1–2 is claimed to have been fully formalized. Appendix D gives representative checks and their boundaries.

Several candidates depend on deep mathematics. The highest-priority interfaces are

the full equation-solution construction in F34/F38(a), graded shortening in F38(c), rational support and integral lifting in N5, the all-rank structural argument in N8, the arbitrary-definable-set input in F41, the general ordered-tree collapse in GA3 and the fixed Diophantine encoding in N9. Finite examples cannot verify these implications.

### 3.3 The deadline and the public record

The last pre-deadline research commit is [401213c](#). The original cutoff was 30 September at 10:04:49.670358 UTC. The unchanged record was observed at 10:05:47.903924 UTC and the final reports and integrity inventories were then prepared administratively. That later work did not extend the research window or add mathematical results. The public handoff commit [0d6b855](#) contains those reports. Paper links to research artifacts use this fixed commit, whose research content agrees with the deadline assessment.

There are 786 terminal process receipts: 589 process successes and 197 nonsuccesses, including 46 timeout flags and 24 interruption flags. They include setup, source rendering, inventories and one deliberately rejected premature final-report command as well as mathematical jobs. They are not counts of proofs or independent tests. The recorded reservation peaks are eight CPU slots, 52 decimal GB in requested per-process limits and seven simultaneous jobs. No recorded budget excess or CPU-allocation conflict was found. These are reservations, not measurements of total memory or consumed CPU time; model inference and unrecorded external tools are outside the accounting.

The final inventory reconciled 31,406 file bindings across 212 selected records, including exact archived versions, without unresolved content. It checked all four frozen launch inputs. Process observations found no remaining attributable worker. These checks establish the integrity of the recorded evidence, not its mathematical correctness. The full [final report](#) and [reproduction guide](#) give the precise limitations. No independent specialist review had been completed when this version was prepared.

## 4 Explicit constructions and obstructions

### 4.1 A virtual isomorphism with no invariant subgroup (F28)

**Problem F28.** *S. Sidki.*

Must an isomorphism from an index-two subgroup of  $F_2$  to a subgroup of  $F_2$  preserve a nontrivial subgroup of its domain? The following example excludes even forward invariance, without a normality or finite-generation assumption on the subgroup.

**Proposition 4.1.** *In  $F = F(a, b)$  let  $S = \langle a, bab^{-1}, b^2 \rangle$  and define*

$$f(a) = b^{-2}, \quad f(bab^{-1}) = b^{-1}a^2b, \quad f(b^2) = b^{-1}a.$$

*Then  $S$  and  $f(S)$  have index two,  $f$  is injective, and  $H \leq S$ ,  $f(H) \leq H$  imply  $H = 1$ .*

*Proof.* The displayed generators of  $S$  are the Schreier basis for the kernel of  $b$ -parity, using transversal  $\{1, b\}$ . Represent  $a, b$  in  $\text{PSL}_2(\mathbb{R})$  by

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Ping-pong on  $\{|x| > 1\} \cup \{\infty\}$  and  $\{|x| < 1\}$  shows that this representation is faithful: every nonzero power of  $A$  sends the second set into the first, and every nonzero power of  $B$  sends the first into the second. Put

$$Q = \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$QAQ^{-1} = B^{-2}, \quad Q(BAB^{-1})Q^{-1} = B^{-1}A^2B, \quad QB^2Q^{-1} = -B^{-1}A.$$

The sign disappears projectively, so  $f$  is injective. For the image index, put  $c = b^{-1}a$ . The image is  $\langle b^2, c, bcb^{-1} \rangle$ , an index-two Schreier subgroup in the basis  $b, c$ .

If  $h \in H$  has an integral determinant-one representative  $M$ , every  $M_n = Q^n M Q^{-n}$  is integral, since it represents  $f^n(h)$  up to sign. For  $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  define

$$E(X) = (8a - 4b + 2c - d)^2 + 7(4b + d)^2 + 7(2c - d)^2 + 49d^2.$$

Expansion gives  $E(QXQ^{-1}) = E(X)$ . The four linear forms determine  $X$ , so the integral matrices of a fixed energy form a finite set. Hence  $M_i = M_j$  for some  $i < j$ , and  $M$  commutes with  $Q^{j-i}$ .

Write  $Q^n = u_n Q + v_n I$ . From  $Q^2 = Q - 2I$  one obtains

$$(u_0, v_0) = (0, 1), \quad (u_{n+1}, v_{n+1}) = (u_n + v_n, -2u_n).$$

For  $n \geq 1$ ,  $u_n$  is odd and  $v_n$  is even. Thus  $u_n \neq 0$ , and commuting with a positive power of  $Q$  implies commuting with  $Q$ . The latter equation gives  $c = -2b$ ,  $d = a - b$ , whence

$$1 = \det M = a^2 - ab + 2b^2 = (a - b/2)^2 + 7b^2/4.$$

Integrality forces  $b = c = 0$ ,  $a = d = \pm 1$ . Thus  $h = 1$  projectively and, by faithfulness, in  $F$ .  $\square$

Related rational conjugations appear in work on self-similar groups [17]. Simplicity there excludes normal invariant subgroups; Proposition 4.1 concerns every subgroup. The integral-matrix implication has a Lean proof, while the free-group interpretation remains a written argument.

*Record.* [Argument and supporting audit for F28.](#)

## 4.2 Short presentations and long Dehn output (H4)

**Problem H4.** *A. Miasnikov.*

Can a finite presentation of a hyperbolic group be converted to a Dehn presentation in polynomial time? We use explicit words in generators and their inverses, and strict length-decreasing Dehn rules.

**Proposition 4.2.** *No uniform polynomial-time algorithm performs this conversion on all finite hyperbolic-group presentations. This remains so when the inputs are infinite non-elementary virtually free groups and the output may use different generators.*

*Proof.* Let a Dehn presentation have  $r$  generators and longest relator length  $L \geq 1$ . A classical observation bounds the number  $T(G)$  of torsion conjugacy classes, including the identity, by

$$T(G) \leq \sum_{j=0}^{L-1} (2r)^j < (2r)^L. \quad (1)$$

Indeed, take a shortest word  $w$  in a nontrivial torsion conjugacy class. It is cyclically reduced. If  $|w| \geq L$ , a Dehn-shortenable subword of an identity power  $w^e$  has length at most  $L$  and lies in a cyclic rotation of  $w$ . Shortening it contradicts conjugacy minimality. This is the usual torsion-conjugacy argument [2, Theorem 3.27]. The same reasoning applies to an explicit finite list of shortening rules, using the largest left-hand-side length.

For  $n \geq 1$  take generators  $x, t_0, \dots, t_n$  and relators

$$t_i = t_{i-1}^2 \ (1 \leq i \leq n), \quad t_n = 1, \quad t_0 x t_0^{-1} = x^2. \quad (2)$$

Put  $Q_n = 2^n$ ,  $M_n = 2^{Q_n} - 1$ . Eliminating the  $t_i$  gives

$$K_n \cong C_{M_n} \rtimes C_{Q_n}, \quad t_0 x t_0^{-1} = x^2.$$

To check equality rather than just a quotient, the relations give  $t_0^{Q_n} = 1$  and  $x^{M_n} = 1$  and collect every word to  $x^i t_0^j$ , so the order is at most  $M_n Q_n$ . The indicated semidirect product realizes that order and the presentation. Multiplication by two has order  $Q_n$  modulo  $M_n$ , since  $0 < 2^j - 1 < M_n$  for  $0 < j < Q_n$ .

The conjugacy classes in  $\langle x \rangle$  are the orbits of this cyclic action, so there are at least

$$M_n / Q_n > 2^{2^n - n - 1}$$

of them. They remain distinct in  $H_n = K_n * \langle z \rangle$  by free-product normal forms. The kernel of the retraction  $H_n \rightarrow K_n$  has free basis  $\{k z k^{-1} : k \in K_n\}$  and finite index  $|K_n|$ . Its rank is at least six. Hence  $H_n$  is infinite, non-elementary and virtually free, in particular hyperbolic.

Adding  $z$  to (2) gives explicit generator-letter size  $4n + 9$ , or  $O(n \log n)$  bits with indexed generators. Any explicit Dehn output of total size  $m_n$  has at most  $m_n$  generators and longest rule length at most  $m_n$ . Therefore

$$m_n \log_2(2m_n) > 2^n - n - 1, \quad m_n > \frac{2^n - n - 1}{n + 1} \quad (n \geq 2).$$

The second inequality follows by separating  $m_n \leq 2^n$  and  $m_n > 2^n$ . Merely writing the output takes superpolynomial time in the input length, regardless of its chosen generators.  $\square$

The torsion hypothesis and explicit-output convention matter. This argument makes no claim about torsion-free input or compressed output. The original finite-input argument, the classical torsion lemma and related earlier output-size observations are separately credited in the [audit](#).

*Record.* [Argument and supporting audit for H4.](#)

### 4.3 Primitive elements in free metabelian groups (M0)

**Problem M0.** *E. I. Timoshenko and V. Shpilrain.*

The entry asks the analogue of F3 for  $M_r = F_r / F_r''$ : does preservation of all primitive elements force an endomorphism to be an automorphism? Primitive means membership in a free basis of the metabelian variety.

**Proposition 4.3.** *For every finite rank  $r$ , an endomorphism of  $M_r$  preserving primitive elements is an automorphism. It suffices to preserve the images of primitive words of  $F_r$ .*

*Proof.* Use the Laurent ring  $R = \mathbb{Z}[X_1^{\pm 1}, \dots, X_r^{\pm 1}]$  and abelianized left Fox derivative columns  $D(w)$ . Write  $J_\phi$  for the matrix with columns  $D(\phi(x_j))$  and  $\lambda = (X_1 - 1, \dots, X_r - 1)$ . The standard identities are

$$D(uv) = D(u) + \bar{u}D(v), \quad \lambda D(w) = \bar{w} - 1, \quad D(\phi(w)) = J_\phi \bar{\phi}(D(w)).$$

Bachmuth's inverse-function criterion, in this convention, says that  $\phi$  is invertible exactly when  $J_\phi$  is invertible over  $R$  [14, Lemma 1]. A primitive word's derivative cannot specialize to zero over a field, since it is a column of an invertible basis Jacobian.

The induced integer matrix on abelianization preserves primitive vectors and is therefore unimodular. Otherwise reduction modulo a prime gives a nonzero kernel vector with a primitive integral lift, obtained by lifting elementary matrices in  $\mathrm{SL}_r(\mathbb{F}_p)$ ; its image is divisible by  $p$ . Compose with a lifted inverse basis change to assume  $\phi$  is IA. Then

$$\lambda J_\phi = \lambda, \quad J_\phi(1, \dots, 1) = I. \quad (3)$$

Ranks zero and one are immediate, so assume  $r \geq 2$ .

If  $\Delta = \det J_\phi$  is not a Laurent unit, augmentation one implies it has at least two monomials. Choose a prime preserving two nonzero coefficients and a maximal ideal containing its reduction. Its residue field  $K$  is finite. The images  $z_i$  of  $X_i$  are nonzero, generate  $K$  as a field, and are not all one. The associated character has a nontrivial finite cyclic image. A Nielsen basis change puts it in the form

$$\chi(x_1) = \dots = \chi(x_{r-1}) = 1, \quad \chi(x_r) = t \neq 1, \quad K = \mathbb{F}_p(t).$$

Indeed, apply integer elementary operations to an exponent row for a generator of that cyclic image; its gcd remains coprime to the image order. The Fox basis change conjugates  $J_\phi(\chi)$ , because  $\phi$  is IA, so singularity is preserved. Equation (3) now puts its kernel in  $W = \mathrm{span}_K(e_1, \dots, e_{r-1})$ .

For distinct  $i, j < r$  and  $P(T) = \sum a_k T^k \in \mathbb{Z}[T]$ , fix the other generators and set

$$\theta(x_i) = x_i \prod_{k \geq 0} x_r^k x_j^{a_k} x_r^{-k}. \quad (4)$$

This is an actual free-group automorphism preserving  $\chi$ , and  $J_\theta(\chi) = I + P(t)E_{ji}$ . Since every element of  $K$  is  $P(t)$ , for  $r \geq 3$  these maps realize  $\mathrm{SL}_{r-1}(K)$  on  $W$ . Its transitivity on nonzero vectors shows that every nonzero kernel vector is  $D(w)(\chi)$  for a primitive word  $w$ . If  $r = 2$ , the nonzero kernel is the line  $Ke_1$ , and  $w = x_1$  suffices. The chain rule gives  $D(\phi(w))(\chi) = 0$ , contradicting primitivity. Thus  $\Delta$  is a unit and the inverse-function criterion applies.  $\square$

The low-rank answers are prior; the all-rank finite-field detection is the proposed additional step. The proof neither assumes that every metabelian automorphism is tame nor substitutes preservation of whole primitive systems for preservation of individual elements. Exact finite-field witnesses include actual free bases and inverses. A later constructor removes a search cutoff by an explicit Kronecker bound; it gives termination without a useful complexity estimate.

*Record.* [Argument and supporting audit for M0.](#)

## 5 Free groups, equation languages and tree actions

### 5.1 Growth of a nonprimitive automorphic orbit (F41)

**Problem F41.** *D. Puder and C. Wu.*

For a nonprimitive word in  $F_r$ , is the automorphic orbit's exponential growth rate  $\sqrt{2r-1}$ ? The identity is an exception to the literal wording: its orbit has ball growth one. We use ordinary length in a fixed basis; this is not a stronger word-dependent cyclic-orbit conjecture.

**Proposition 5.1.** *Let  $r \geq 2$ ,  $q = 2r - 1$ , and let  $w \neq 1$  be nonprimitive in  $F_r$ . There are constants  $C_w > 0$ ,  $D_w \geq 0$  such that*

$$|\mathrm{Aut}(F_r)_w \cap B(n)| \leq C_w(n+1)^{D_w} q^{n/2}.$$

*The ball growth limit and the spherical growth limsup are  $\sqrt{q}$ .*

Two imported results are essential. Kharlampovich–Myasnikov’s arbitrary-definable-set theorem says that a definable subset of  $F_r$  or its complement is a sub-multipattern [18, Definitions 5–6, Theorem 13]. A one-variable pattern is described by finitely many noncancelling equations with fixed coefficients; every variable in its first equation is a repeated piece of the output, possibly with inverse orientation and overlap. A sub-multipattern is a subset of a finite union of these patterns. Pillay’s parameter-free generic type  $p_0$  is realized by basis elements, and its realizations in finite-rank free groups are primitive [25, Fact 1.10, Theorem 2.1]. Every parameter-free definable set containing a generic element has finitely many left translates covering the whole group.

*Derivation of Proposition 5.1.* First count reduced words with a fixed cover by repeated pieces. Suppose a first pattern equation has  $t$  variable blocks and  $C$  coefficient letters. For length  $n$ , choose the block lengths and a second occurrence of each block, including orientation. There are at most a constant times  $(n + 1)^{2t}$  choices. In each choice form the signed graph of identified letter positions. An inconsistent inverse cycle makes the choice impossible. In a consistent choice every noncoefficient position is joined to a different position, so the number  $c$  of components satisfies  $c \leq (n + C)/2$ .

Collapse these components in the path of consecutive positions. The quotient is connected. On a spanning tree, the root component has  $2r$  possible letters, while each further component has at most  $q$ : one letter is excluded by free reduction across its parent edge, with the prescribed signs. Ignoring other constraints only enlarges the count. Thus each choice contributes at most  $2rq^{(n+C)/2}$  words. Finite unions and subsets preserve a polynomial times  $q^{n/2}$  bound. No sub-multipattern is finite-translate generic, since translating by a fixed element changes the radius by at most a constant, whereas  $F_r$  has ball growth  $q$ .

Since  $w$  is nonprimitive, it does not realize  $p_0$ . Choose a parameter-free formula  $\varphi \in p_0$  false at  $w$ . Its negation defines an automorphism-invariant set  $P$  containing the orbit of  $w$ . The complement contains a basis element and is finite-translate generic. The definable-set dichotomy and the preceding count therefore make  $P$ , not its complement, a sub-multipattern. This proves the upper bound. No definability of the orbit or of the primitive set is assumed.

For the lower bound, replace  $w$  by a cyclically reduced conjugate  $v$ , of length  $\ell$ . There are at least  $(2r - 2)q^{m-1}$  reduced words  $t$  of length  $m$  for which  $tvt^{-1}$  is reduced as written: exclude the inverse of the first letter of  $v$  and the last letter of  $v$  from the last letter of  $t$ . Distinct  $t$  give distinct words of length  $2m + \ell$ . This gives the ball lower bound and the matching spherical limsup. Empty parity classes can prevent a spherical limit.  $\square$

In rank one every orbit is finite. The higher-rank proof is a proposed consequence of the two deep cited theorems and the piece-cover count; its finite checks do not verify the model-theoretic interface.

*Record.* [Argument and supporting audit for F41.](#)

## 5.2 Discriminating a free square (GA3)

**Problem GA3.** *A. Miasnikov.*

If  $G$  acts freely on a  $\Lambda$ -tree, can each finite set of nonidentity elements of  $G * G$  survive a homomorphism to  $G$ ? For nontrivial abelian  $G$  the literal answer is negative: a commutator between the two factors always dies. Free actions below have no inversions.

**Proposition 5.2.** *If  $G$  is nonabelian and  $\Lambda$ -free, for each finite  $W \subset (G * G) \setminus \{1\}$  there are  $h \in G$  and  $N$  such that the maps which are the identity on the first factor and  $b \mapsto h^n b h^{-n}$  on the second are nontrivial on  $W$  for all  $n \geq N$ . No finite-generation hypothesis on  $G$  is required.*

*Derivation.* It suffices first to control a finitely generated nonabelian subgroup  $K$ . In a divisible scalar extension of its free  $\Lambda$ -tree, choose a finite symmetric generating set  $S$  and the scale

$$L = \max\{\ell(s), \ell(st) : s, t \in S\} > 0.$$

The disjoint-axis formula bounds pairwise distances between generator axes by  $L/2$ . The finite Helly property gives a point within  $L$  of every axis, hence with generator displacements at most  $3L$ . The hull of its orbit has distances in the principal convex subgroup generated by  $L$ . Quotient by its maximal infinitesimal convex subgroup and extend the resulting Archimedean ordered group to  $\mathbb{R}$ . An element attaining translation length  $L$  stays hyperbolic, so this collapse does not have a global fixed point.

Guirardel's collapse theorem and its general- $\Lambda$  remark give abelian arc fixators and trivial tripod fixators [13, Fact 5.1]. The final scalar extension preserves these properties because the divisible Archimedean subgroup is dense and no branch points are added. The CSA property [13, Lemma 2.5] implies that a nonabelian  $K$  has no nontrivial abelian normal subgroup: such a subgroup would lie in every conjugate of its maximal abelian subgroup, contradicting malnormality. Thus  $K$  preserves no line and fixes no end. For a line, the arc kernel is abelian and the remaining faithful torsion-free action is by translations. For an end, the Busemann kernel is abelian because any two of its elements fix a common terminal ray. Either case would make  $K$  abelian.

In this real-tree action each nonidentity element fixes at most two ends, since three fixed ends give a fixed tripod. Given finitely many nonidentity coefficients  $C \subset K$ , choose two limit-set ends whose pair is disjoint from all its  $C$ -translates. Tree ping-pong, or the elementary separator in the retained supplement, gives a hyperbolic  $h$  with both ends in small prescribed neighborhoods  $U_+, U_-$  and

$$c(U_+ \cup U_-) \cap (U_+ \cup U_-) = \emptyset \quad (c \in C).$$

Sufficient powers send the complement of  $U_-$  into  $U_+$  and the complement of  $U_+$  into  $U_-$  in the reverse direction. This is the standard boundary separation mechanism; its prior use is credited in the [source audit](#).

Cyclically reduce each word in  $W$ . A core in one free factor survives automatically. Write every other core as  $a_1 b_1 \cdots a_m b_m$ , and include all coefficients, together with two noncommuting elements of  $G$ , in the generators of  $K$ . Starting with an end in  $U_-$ , read the substituted core from right to left. Each  $h^{-n}$ ,  $b_i$ ,  $h^n$ ,  $a_i$  sends it successively to  $U_-$ , outside both neighborhoods,  $U_+$  and outside both. Its final image is outside  $U_-$ , so the word is nonidentity. The same  $h, N$  work for all cores. The factor maps are automorphisms of all of  $G$ , which extends the argument from  $K * K$  to  $G * G$ .  $\square$

The trivial group satisfies the property vacuously. This separates the literal abelian obstruction from the substantive nonabelian assertion. The arbitrary- $\Lambda$  collapse, not CSA alone, is the principal structural dependency. No procedure for obtaining  $h, N$  from an unspecified presentation is asserted.

*Record.* [Argument and supporting audit for GA3.](#)

### 5.3 A shared polynomial-identity method

Let  $F_r$  have basis  $A$ , and let  $B_\phi$  be the exponent-sum matrix of an endomorphism. If  $\det B_\phi \neq 0$ , then  $\phi$  is injective: its image is a free group generated by  $r$  elements, and its ambient abelianization has rank  $r$ , so its rank is exactly  $r$ ; Hopficity then applies to the surjection onto that image.

We use the effective *full solution relation* of word equations with involution and regular constraints [9, Theorem 4.3]. It has the form

$$\{(h(d_1), \dots, h(d_k)) : h \in \mathcal{R}\}, \quad (5)$$

where  $\mathcal{R}$  is a regular control of explicit endomorphisms of one finite extended alphabet, with separately designated output letters  $d_i$ . Group equations reduce to these monoid equations by triangulation. For reduced  $X, Y, Z$ , a triangle  $XY = Z$  is replaced by

$$X = ST, \quad Y = \bar{T}R, \quad Z = SR,$$

with reduced-word constraints; maximal cancellation proves equality of the projected solution relations. Cyclic reduction and positivity are regular constraints in the prescribed basis. Recognizing monoids can be made compatible with involution using  $M \times M^{\text{op}}$ .

**Lemma 5.3.** *One can decide whether a rational polynomial in the separate output letter counts vanishes on every tuple of (5).*

*Proof.* Incidence matrices act linearly on the stacked count vector  $y \in \mathbb{N}^D$ . For polynomial degree  $d$ , lift  $y$  to the vector  $\nu_d(y)$  of all monomials of degree at most  $d$ , including one. Its dimension is  $\binom{D+d}{d}$ . Each transition induces an effective rational matrix  $T^{[d]}$  satisfying  $\nu_d(Ty) = T^{[d]}\nu_d(y)$ . In each automaton state compute the rational span of reachable lifted vectors. Propagating these spaces can strictly increase total dimension only finitely many times. The stable spaces are exactly the spans of actual path vectors, by induction in both directions. Test the linear functional representing the polynomial on all accepting spaces. Reverse the control automaton if required by morphism-composition order. If terminal filtering is not built into the chosen convention, include the finite support of the count vector in the control state. This also handles erasing morphisms.  $\square$

This decides a universal identity, equivalently the existence of a *nonzero* value. It does not decide whether an arbitrary polynomial has a zero among language outputs.

## 5.4 Potential positivity (F34)

**Problem F34.** *A. Miasnikov and V. Shpilrain.*

Part (a) asks whether some automorphism makes a supplied word positive in the fixed basis. Part (b) asks whether enlarging the ambient free rank creates new potentially positive words; its negative answer is prior [5].

**Proposition 5.4.** *Potential positivity is decidable uniformly in finite rank.*

*Proof.* An injection  $\phi : F_r \rightarrow F_s$  reflects potential positivity. Indeed, orient every edge of the based Stallings graph of  $\phi(F_r)$  by its positive ambient label. Choose a spanning tree and give each non-tree edge this orientation in the resulting free basis. A positive closed path stays positive when the tree is collapsed, so  $\phi(w)$  positive implies  $w$  positive in a basis of its domain. No finite-index assumption is needed.

Consequently  $w$  is potentially positive exactly when there is an endomorphism with  $\phi(w)$  positive and  $\det B_\phi \neq 0$ . Encode  $w(X_1, \dots, X_r) = V$  with  $V \in A^*$  and construct its full relation (5). The determinant is a polynomial in the separate counts of the  $X_j$ . Lemma 5.3 decides whether it is identically zero. A nonzero output gives an injection and the graph argument supplies a witnessing automorphism. The identity is positive under the site's non-negative-letter convention; an alternative nonempty convention changes only that boundary case.  $\square$

Rank-two decisions and the stability answer are credited to earlier work in the [scope audit](#). The complete equation-to-language construction is imported and was not implemented. The graph and polynomial stages and the language interfaces were implemented and checked; that is a narrower assertion.

*Record.* [Argument and supporting audit for F34.](#)

## 5.5 Translation equivalence (F38)

**Problem F38.** *I. Kapovich and P. Schupp.*

Part (a) asks whether  $\|\alpha(u)\| = \|\alpha(v)\|$  for all automorphisms can be decided. Part (c) replaces equality by two-sided bounds on their ratio, uniform in the ambient automorphism. Part (b), a substitution symmetry for translation-equivalent pairs, is already answered affirmatively by Lee [21].

**Proposition 5.5.** *Translation equivalence is decidable uniformly in finite rank.*

*Proof.* Kapovich–Levitt–Schupp–Shpilrain identify equality under all automorphisms with equality under all injective endomorphisms [16, Corollary 1.4]. Thus the question is equivalent to

$$\det B_\phi(\|\phi(u)\| - \|\phi(v)\|) = 0 \quad \text{for every endomorphism } \phi.$$

Nonzero determinant guarantees injectivity; conversely every automorphism has determinant  $\pm 1$ . Encode  $P^{-1}u(X)P = U$  and  $Q^{-1}v(X)Q = V$ , with  $U, V$  cyclically reduced, allowing the empty word. Every endomorphism extends to such a tuple, and  $|U|, |V|$  are exactly the two cyclic lengths. The displayed expression is a polynomial of degree  $r + 1$  in separate letter counts. Apply Lemma 5.3 to the full solution relation.  $\square$

For bounded equivalence write  $H_w \leq \text{Out}(F_r)$  for the stabilizer of the *oriented* conjugacy class of  $w$ .

**Proposition 5.6.** *For nonidentity  $u, v$ , the inequalities  $\|\alpha(v)\| \leq C\|\alpha(u)\|$  for some  $C$  and all  $\alpha \in \text{Aut}(F_r)$  hold exactly when  $H_u[v]$  is finite. Hence bounded translation equivalence is equivalent to commensurability of  $H_u, H_v$ , and is decidable.*

Necessity follows because bounded cyclic length leaves finitely many conjugacy classes. The proposed sufficiency uses a quantitative relative shortening lemma, proved from Sela’s *graded* shortening theorem, with a separate treatment of primitive powers in rank two. The exact reduction and its normalization are in Appendix B. An unqualified replacement of automorphisms by arbitrary injections would be false for this bounded question.

The finite decision step is more elementary once the criterion is granted. Whitehead peak reduction computes generators for  $H_u, H_v$ . Let  $I = \ker(\text{Out}(F_r) \rightarrow \text{GL}(r, \mathbb{F}_3))$ . Handel–Mosher’s aperiodicity theorem says a periodic conjugacy class under an element of  $I$  is fixed [15, Theorem 4.1]. For a finitely generated  $H$ , the orbit  $H[w]$  is finite exactly when  $H \cap I$  fixes  $[w]$ . Explore the finite mod-three image, retaining actual automorphism representatives and the image of  $[w]$  at each matrix. Different classes at the same matrix certify an element of  $H \cap I$  moving  $[w]$  and therefore an infinite orbit. Otherwise the finite graph closes consistently. Apply this to both orbits.

The code implements that finite procedure; it does not calculate a shortening constant or validate the graded theorem. Rank-two decisions are prior [23, 22]; the proposed additional range is higher rank. For identity inputs, the ratio is undefined; its natural two-sided inequality extension is true for two identities and false for exactly one. Rank one is immediate.

*Record.* [Argument and supporting audit for F38.](#)

## 6 Decisions and undecidability in nilpotent groups

### 6.1 Direct decomposition with torsion (N5)

**Problem N5.** *G. Baumslag.*

Is direct indecomposability decidable for finitely generated nilpotent groups? The torsion-free case is prior [3]. Here the input is a finite presentation *promised to define a nilpotent group*; no recognition algorithm for arbitrary presentations is claimed.

**Proposition 6.1.** *Under this promise there is a uniform algorithm deciding whether  $G = A \times B$  with both factors nontrivial, and constructing generators for such factors when they exist.*

*Proof.* Use standard effective nilpotent-group operations: consistent polycyclic presentations, subgroup membership, centres, finite torsion subgroups, kernels and intersections. Let  $T = T(G)$ ,  $Z = Z(G)$ ,  $Q = G/T$ , and let  $R$  be the rational Malcev completion of  $Q$ . Rational direct decomposition is effective. Uniqueness modulo the centre of indecomposable rational factors [11, Theorem 3.3], also used in [3], says that any decomposition matches a partition of a fixed decomposition  $R = R_1 \times \cdots \times R_m$ , modulo  $Z(R)$ .

Put  $K = G/Z$  and let  $\kappa : K \rightarrow R/Z(R)$  be induced by the natural map. Its kernel is  $S = T(K)$ . To see this, the preimage  $C$  in  $G$  of  $Z(Q) = Q \cap Z(R)$  satisfies  $\ker \kappa = C/Z$ . Choose a finite quotient injective on  $T$ , using residual finiteness, and let  $e$  be its exponent. The characteristic power subgroup  $G^e$  is torsion-free and of finite index. For  $c \in C$ ,  $[c^e, g]$  lies in both  $T$  and  $G^e$  for every  $g$ , so  $c^e \in Z$ . Conversely a central power has central image in torsion-free  $Q$ , by unique roots. Thus  $C/Z = T(K)$ , which is finite and computable.

For each ordered partition  $I, J$  of the rational factors compute

$$K_1 = \kappa^{-1}(R_I Z(R)/Z(R)), \quad K_2 = \kappa^{-1}(R_J Z(R)/Z(R)).$$

If  $G = A \times B$  induces this partition, its images  $Y_1, Y_2$  in  $K$  form a direct product and satisfy  $K_i = Y_i S$ . Hence  $[K_i : Y_i] \leq |S|$ . Enumerate all subgroups of  $K_i$  up to that index, using finite permutation representations, and retain exactly the pairs with

$$K = Y_1 Y_2, \quad [Y_1, Y_2] = 1, \quad Y_1 \cap Y_2 = 1.$$

This is a finite complete list of quotient decompositions that might lift. Empty partitions and trivial  $Y_i$  are kept, so central and finite factors are not lost.

Let  $X_i$  be the full preimages in  $G$ . Discard a pair if  $[X_1, X_2] \neq 1$ . Otherwise  $X_1 X_2 = G$  and  $X_1 \cap X_2 = Z$ . Choose presentations of  $Y_i$  and lift their generators to  $X_i$ . The lifted relators are central defects  $v_i \in Z^{m_i}$ ; changing lifts by central elements  $z$  changes them to  $v_i + E_i z$ , where  $E_i$  is the exponent-sum matrix. For a proposed splitting  $Z = Z_1 \oplus Z_2$ , the extension  $X_i/Z_i \rightarrow Y_i$  has a section exactly when  $E_i z = -v_i$  is solvable over  $Z/Z_i$ .

Smith reduction converts this into finitely many conditions

$$v \in Z_i + dZ, \quad d \geq 0, \tag{6}$$

where  $d = 0$  means exact membership. Appendix A gives a finite constructive decision for these conditions over an arbitrary finitely generated abelian group, retaining torsion, integral splittings and required nontrivial factors. Corrected lifts and  $Z_i$  then generate factors  $A_i$ . They commute, intersect trivially and generate  $G$ . Every actual decomposition occurred in the finite quotient list and passes precisely these lifting conditions, proving completeness and termination.  $\square$

The rational decomposition machinery is credited prior work; the proposed additional scope is torsion. In particular, rational decomposability alone does not decide integral decomposability. The final implementation accepts promised-nilpotent finite presentations without a supplied class bound and returns original-generator factor words. Its negative integral-gluing controls and finite comparisons support implementation details, not the universal support argument.

*Record.* [Argument and supporting audit for N5.](#)

## 6.2 Single commutators in every free nilpotent group (N8)

**Problem N8.** *A. Miasnikov.*

Part (a) concerns  $[x, y] = g$  in general finitely generated class-two groups and has a prior negative answer [27]. Part (b) asks the same question in every finitely generated free nilpotent group. We use  $[x, y] = x^{-1}y^{-1}xy$  throughout this section.

**Proposition 6.2.** *Given finite rank  $r$ , class  $c$  and a word  $g$  in  $N_{r,c} = F_r/\gamma_{c+1}(F_r)$ , the equation  $[x, y] = g$  is decidable; on a positive answer its factors can be constructed.*

The main issue is the finite treatment of correction parameters, not enumeration of positive witnesses. Here is the proposed algorithm's structure, with its separation argument detailed in Appendix C.

For  $g \neq 1$ , let  $d$  be its first nonzero lower-central degree and  $W$  its leading integral Lie coordinate. Exact Nielsen operations on the pair preserve its commutator. They normalize every solution to leading terms

$$C \in (L_{\mathbb{Z}})_p, \quad D \in (L_{\mathbb{Z}})_q, \quad [C, D] = W, \quad p \leq q, \quad p + q = d.$$

Here  $L_{\mathbb{Z}}$  is the untruncated free integral Lie ring. A finite list of leading pairs suffices. For unequal weights, the projective bracket map has finite fibres; rational points and all signed integral scale divisors are enumerated. For equal weights, the exterior tensor gives a rank-two lattice, and finitely many Hermite representatives suffice modulo the exact  $\mathrm{SL}_2(\mathbb{Z})$  operations. Primitive normalization alone would discard valid solutions.

At offset  $s$  the new coordinate map is

$$A_s(U, V) = [U, D] + [C, V]. \tag{7}$$

The structural kernel analysis separates two cases. After the offset  $q - p$ , its elements lie in  $\mathrm{Lie}(C, D)$  and have exact full-rank integral period lattices, obtained by denominator clearing in rational commutator-preserving substitutions. Every residue is retained. Before that offset a nonzero kernel has dimension at most one; these are the exceptional parameters.

At the first exceptional offset  $t$ , solve *all* equations from  $t$  through  $2t - 1$  as one affine integer system. If the remaining class ends before  $2t$ , all remaining equations are jointly linear and the branch terminates by integer linear algebra. Otherwise choose an integral parameter  $T$  for the first exceptional coordinate in the complete surviving lattice. Its image can have a nonunit step, which must be retained. At offset  $2t$  there is a scalar quadratic obstruction in  $T$  whose quadratic coefficient survives after quotienting by both the new correction image and *all later parameter columns*. It has at most two integral roots. Fix the prefix through  $t$  at each root and restart, leaving higher coordinates free again.

The diagonal positional-polynomial lemma in Appendix C is the reason later parameters cannot cancel this quadratic term. Each branch has finitely many children and every restart fixes a higher offset, so the recursion terminates. An assembled solution is checked in the

actual group. Completeness follows by retaining every leading branch, every exact-period residue and every integral root forced by the full correction block.

The classical inputs include Hall/Malcev coordinates, Shirshov–Witt, Lazard elimination and the two-coefficient inner-solution theorem for free Lie algebras [1, 26]. They are applied in suitable free subalgebras *before* truncation. The original free class-two case is prior; the proposed contribution is the general finite recursion. Its full word-input implementation exists, with positive and negative native GAP reconstructions, but the all-rank structural proof remains a major review priority.

*Record.* [Argument and supporting audit for N8.](#)

### 6.3 One fixed group with an undecidable retract problem (N9)

**Problem N9.** *A. Miasnikov.*

Part (a) asks whether the retract problem is decidable in every finitely generated nilpotent group. Its background distinguishes one fixed ambient group from the previously established class-wide undecidability [27]. Part (b), for free nilpotent groups, has a prior positive answer [28].

**Proposition 6.3.** *There is a fixed finitely generated torsion-free class-two group  $G$  with undecidable retract problem, already on a computable family*

$$H_n = \langle x_0, x_1^{3n+1} x_2^{-1} \rangle \quad (n \in \mathbb{N})$$

*of isolated rank-two free class-two subgroups. Their images in both lower-central layers are primitive sublattices.*

*Proof.* Choose a nonrecursive computably enumerable set  $S \subset \mathbb{N}$  containing 0, 1. DPRM gives one fixed integer polynomial  $P$  such that

$$n \in S \iff \exists y_1, \dots, y_m \in \mathbb{Z} \ P(n, y) = 0.$$

The integer version follows by replacing natural-number existential variables by sums of four squares [20]. Fix an arithmetic circuit using additions, multiplications and integer constants. Let  $p, o$  be its parameter and output labels.

Use indices 0, 1, 2, 3 and two new indices  $U_z, V_z$  for each scalar label  $z$ . Let  $L$  be the full integer lattice of alternating matrices  $M$  on these indices satisfying the following *linear* equations:

$$\begin{aligned} M_{0,U_z} &= M_{1,V_z} = 0, & M_{1,U_z} + M_{0,V_z} &= 0, \\ M_{0,V_z} &= cM_{01} & & \text{for a constant } z = c, \\ M_{0,V_z} &= M_{0,V_x} + M_{0,V_y} & & \text{for } z = x + y, \\ M_{U_x,V_y} &= M_{0,V_z} & & \text{for } z = xy, \\ M_{0,V_o} &= 0, \quad M_{03} = M_{12} = 0, & M_{13} &= M_{02} = 3M_{0,V_p}. \end{aligned}$$

Multiplication is encoded by imposing rank two on  $M$ , not by nonlinear equations defining the lattice.

We claim that a rank-two  $M \in L$  satisfying

$$(3n+1)M_{01} - M_{02} = 1 \tag{8}$$

exists precisely when  $P(n, y) = 0$  is integrally solvable. Put  $d = M_{01}$ ,  $x = M_{02}$  and  $t = 3n+1$ . The Pfaffian on 0, 1, 2, 3 gives  $dM_{23} = x^2$ , while  $td - x = 1$ . Hence

$$d(M_{23} - t^2d + 2t) = 1.$$

Thus  $d = \pm 1$ ; since  $3 \mid x$ , the negative sign is impossible. So  $d = 1$ ,  $x = 3n$  and  $M_{0,V_p} = n$ . Define  $u_i = M_{i1}$ ,  $v_i = M_{0i}$ . All rank-two Pfaffians now give

$$M_{ij} = u_i v_j - v_i u_j.$$

The two coordinates for each scalar are  $(u_{U_z}, v_{U_z}) = (z, 0)$  and  $(u_{V_z}, v_{V_z}) = (0, z)$ . Thus the remaining linear equations enforce the circuit over  $\mathbb{Z}$ . Conversely a circuit solution gives these coordinate pairs and

$$(u_0, v_0) = (1, 0), (u_1, v_1) = (0, 1), (u_2, v_2) = (0, 3n), (u_3, v_3) = (-3n, 0),$$

which produce the required matrix.

Compute an integer basis  $B_1, \dots, B_s$  of  $L$  and define  $G$  by

$$[x_i, x_j] = \prod_{\ell=1}^s z_\ell^{B_\ell[i,j]}, \quad [x_i, z_\ell] = [z_\ell, z_k] = 1. \quad (9)$$

On  $\mathbb{Z}^D \times \mathbb{Z}^s$  its multiplication is

$$(a, c)(b, e) = \left( a + b, c + e - \sum_{i < j} a_j b_i B[i, j] \right).$$

Bilinearity proves associativity and the normal form realizes the presentation exactly, proving torsion-freeness. All data are fixed independently of  $n$ . Put  $a = x_0$ ,  $b_n = x_1^{3n+1} x_2^{-1}$  and  $c_n = [a, b_n]$ , with central coordinate  $q_n = (3n+1)B[0, 1] - B[0, 2]$ . A solution at  $0 \in S$  proves  $q_n \neq 0$  for every  $n$ . Thus  $H_n$  is the integral Heisenberg group with central generator  $c_n$ .

A retraction  $\rho : G \rightarrow H_n$  must send  $z_\ell$  to  $c_n^{\lambda_\ell}$  and  $x_i$  to  $a^{r_i} b_n^{s_i} c_n^{k_i}$ . The commutator relations give

$$M = \sum_{\ell} \lambda_\ell B_\ell, \quad M_{ij} = r_i s_j - s_i r_j.$$

Fixing  $c_n$  gives (8), so  $M$  has rank two and  $n \in S$ . Conversely a rank-two solution of (8) gives a homomorphism by  $x_i \mapsto a^{u_i} b_n^{v_i}$ ,  $z_\ell \mapsto c_n^{\lambda_\ell}$ . Its displayed coordinates fix  $a$  and  $b_n$ , so it is a retraction. Therefore  $H_n$  is a retract if and only if  $n \in S$ .

For the asserted isolation,  $L$  is the kernel of an integer matrix, hence saturated. Its basis matrix has an integer right inverse, so the commutator columns generate  $\mathbb{Z}^s = G'$ . Matrices  $M_0, M_1$  from solutions at  $0, 1$  give coefficient vectors  $\mu_0, \mu_1$  with

$$((1-n)\mu_0 + n\mu_1) \cdot q_n = (1-n)(3n+1) + n(3n-2) = 1.$$

Thus  $q_n$  is primitive even when  $n \notin S$ . The abelianized subgroup has basis  $e_0, (3n+1)e_1 - e_2$ , a direct summand. Also  $H_n \cap G' = H'_n = \langle c_n \rangle$ . If  $g^k \in H_n$ , first use the primitive abelianized image to write  $g = hz$  with  $h \in H_n$ ,  $z \in G'$ , then use primitivity of  $q_n$  to conclude  $z \in \langle c_n \rangle$ . Hence  $g \in H_n$ .  $\square$

This is an effective construction from a fixed DPRM polynomial, whose full numerical presentation was not expanded. The implemented small circuit checks and Lean normalization lemma do not verify DPRM or the whole reduction. Primitivity of the graded images alone does not make the subgroup a retract: it does not remove the ambient defining relations.

*Record.* [Argument and supporting audit for N9.](#)

## 7 Two partial results

### 7.1 Effective free-metabelian growth (G9)

**Problem G9.** *Collection entry, with no individual author listed.*

The question asks for the growth rate of a free metabelian group of rank  $r$ , with its standard generators. An approximation algorithm and an interval leave the exact constant undetermined.

Let  $V_r(n)$  be the cardinality of the radius- $n$  ball in  $M_r$  and  $\lambda_r = \lim_n V_r(n)^{1/n}$ . Put

$$P(t) = \#\{s_1 > \cdots > s_k > 0 : \sum s_i \leq t\}, \quad K_n = (n+1)^3 P(n+1)^2,$$

including the empty list in  $P(t)$ .

**Proposition 7.1.** *Uniformly in  $r$  and  $n \geq 1$ ,*

$$\max \left\{ 1, \left( \frac{V_r(n)}{K_n} \right)^{1/(n+2)} \right\} \leq \lambda_r \leq V_r(n)^{1/n}. \quad (10)$$

For these endpoints  $L_n, U_n$ ,

$$\log(U_n/L_n) \leq \frac{2\log(2r+1) + 3\log(n+1) + 4\sqrt{n+1}}{n+2}. \quad (11)$$

Thus  $\lambda_r$  is uniformly computable. The retained rank-two certificates give

$$\boxed{2.676891785 \leq \lambda_2 \leq 2.943737759.}$$

*Proof of the approximation assertion.* A word in  $F_r$  is a path in  $\mathbb{Z}^r$ . Its finitely supported net integral edge flow determines its image in  $M_r$ : this is the classical flow criterion, obtained from the first homology of the abelian cover [12, 24]. In particular  $V_r(n)$  is computable by enumerating words of length at most  $n$  and deduplicating flows. No shortest-word oracle is required.

Use the first coordinate as height. A strict bridge path starts at height zero, is positive thereafter, and ends at its maximum height  $H$ . Its edges, indexed by their upper height, lie in the band  $(0, H]$ . Concatenating bridge flows of equal height places them in disjoint bands. Restricting the product flow to these bands and recovering successive endpoints from boundaries recovers every factor. If  $b(n, H)$  counts bridge elements with a representative of length at most  $n$  and height  $H$ , then

$$b(n, H)^k \leq V_r(kn), \quad b(n) := \sum_{H=1}^n b(n, H) \leq n\lambda_r^n.$$

For each element with a strict half-space representative, choose one deterministically. Cut it at the last highest remaining vertex, then the last lowest, and so on. The positive successive spans strictly decrease and their sum is at most its length. Reflect downward pieces and concatenate them into a bridge. Given its *flow* and the span list, restrictions to disjoint bands recover the original piece flows and their endpoints; reversing reflections and translations recovers the original element. Thus, writing  $h(n)$  for the number of half-space elements,

$$h(n) \leq P(n)b(n) \leq P(n)n\lambda_r^n.$$

This adapts the classical bridge unfolding argument [10]; the additional requirement here is recovery from total flow rather than from a path, allowing repeated edges and cancellations.

Split an arbitrary representative  $w = UV$  of length at most  $n$  at a vertex of globally minimal height, after  $i$  letters. Then  $a_1U^{-1}$  and  $a_1V$  are strict half-space words of lengths at most  $i + 1$  and  $n - i + 1$ , and their quotient recovers  $w$ . Hence

$$V_r(n) \leq \sum_{i=0}^n h(i+1)h(n-i+1) \leq (n+1)^3 P(n+1)^2 \lambda_r^{n+2}.$$

This proves the lower bound. Submultiplicativity of balls gives the upper bound. For  $s > 0$ ,

$$P(t) \leq e^{st} \prod_{j \geq 1} (1 + e^{-sj}) \leq \exp(st + 1/s),$$

so  $P(t) \leq e^{2\sqrt{t}}$ . Combining this with  $V_r(n) \leq (2r+1)^n$  gives (11). The effective error bound and exact finite counts yield rational outward approximations of any requested precision. The procedure need not be practical.  $\square$

The numerical lower bound uses a free monoid of bridge atoms, with infinite completed boundary and shear families. The upper bound uses 648 verified rewriting relations, a 1,968-state avoidance automaton and exact positive-vector inequalities. Appendix D.1 gives the finite certificate structure and the final rational series. The flow-support connection construction used to shorten representatives is prior work [24, Section 2.7]; no new general geodesic algorithm or current best-in-literature bound is asserted. The method's recorded extension to free solvable groups is not another problem count.

*Record.* [Argument and supporting audit for G9.](#)

## 7.2 Infinitely many special five-strand braids (B9)

**Problem B9.** *P. Dehornoy.*

Let  $S(\sigma_i) = \sigma_{i+1}$  and

$$a \triangleright b = aS(b)\sigma_1 S(a)^{-1}.$$

The special braids are the closure of 1 under this operation in  $B_\infty$ . How many lie in each standard subgroup  $B_N$ ? Intermediate terms in a special expression may have more strands than its final value.

**Proposition 7.2.** *For every  $n \geq 3$  the braid*

$$\beta_n = \sigma_1^n \sigma_3^{n-2} \sigma_2^{1-n} \sigma_3^2 \sigma_4^{-1} \sigma_2 \sigma_1 \sigma_3^{n-1} \sigma_4^{2-n} \sigma_2^{-n}$$

*is special, has standard strand support exactly five, and the  $\beta_n$  are pairwise distinct. There are therefore countably infinitely many special braids in every  $B_N$  with  $N \geq 5$ .*

*Proof.* Define  $u_0 = 1$ ,  $u_1 = \sigma_1$ , and  $u_{n+1} = u_n \triangleright u_{n-1}$ . Induction gives  $u_n S(u_{n-1}) = \sigma_1^n$ . For  $n \geq 3$  set

$$q_n = u_{n-3}, \quad A_n = \sigma_1^n \sigma_3^{n-2} \sigma_2^{1-n}.$$

Expanding the recurrence three times and commuting distant generators gives  $u_n S^3(q_n) = A_n$ . Now take the explicitly special expression

$$v_n = 1 \triangleright ((1 \triangleright q_n) \triangleright 1), \quad \beta_n = u_n \triangleright v_n.$$

Expansion yields

$$v_n = S^2(q_n) \sigma_2^2 \sigma_3^{-1} S^3(q_n)^{-1} \sigma_1.$$

In the next operation, move  $S^4(q_n)^{-1}$  past  $\sigma_2\sigma_1$  and apply the recurrence. The result is

$$\beta_n = A_n \sigma_3^2 \sigma_4^{-1} \sigma_2 \sigma_1 S(A_n)^{-1},$$

which is the displayed five-strand word, of length  $6n - 1$  as written and exponent sum three.

Use the unreduced Burau representation at  $t = -1$ , whose nontrivial two-by-two block is

$$T = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = I + N, \quad N^2 = 0, \quad T^k = I + kN \quad (k \in \mathbb{Z}).$$

Multiplying the fifth row through the displayed blocks gives

$$(0, n(n-1), 1 - n^2, -n^2 + 3n - 2, n^2 - 2n + 2).$$

Its second entry is strictly increasing for  $n \geq 3$  and nonzero, whereas every standard four-strand braid has fifth row  $e_5$ . This proves distinctness and strand support; faithfulness is not needed to distinguish unequal images. The same representation extends by identities for  $N \geq 5$ . Countability of  $B_N$  finishes the proof.  $\square$

The small-strand counts 1, 2, 4 follow from Dehornoy's prior structural results [8, Sections 4–5]. In particular a special braid in  $B_N$  has exponent sum between 0 and  $N - 1$ ; sum zero gives 1 and sum  $N - 1$  gives  $\tau_{N-1} = \sigma_{N-1} \cdots \sigma_1$ . The exponent-one elements are exactly  $a \triangleright 1$  with  $a$  special in  $B_{N-1}$ , and this map is injective. Thus

$$\mathcal{C} \cap B_3 = \{1, \sigma_1, \sigma_1^2 \sigma_2^{-1}, \sigma_2 \sigma_1\}.$$

In  $B_4$  only exponent two remains. Writing  $I_p(A) = A\tau_p S(A)^{-1}$ , that sector corresponds to right cosets  $AB_2$  in  $B_3$  admitting  $A = aS(c)$  with both  $a, c$  special. Four known cosets give at least ten special braids in total; exhaustion was not proved.

Later reductions decide the complete partner set for each supplied first braid and exclude several infinite first-input families. For example, for  $I_1(\beta_n)$ ,  $n \geq 3$ , an invariant Burau entry  $n(n-1)$  excludes every partner returning to the required three-strand subgroup. These results do not classify unrestricted first inputs. The global four-strand question therefore remains open here.

*Record.* [Argument and supporting audit for B9.](#)

## 8 Prior results, failures and assessment

### 8.1 What was not counted

Several useful outcomes are outside the twelve-entry tally. Arguments for F11 and F42 were independently derived and then identified with prior answers. Achyuth Jayadevan's prior Lean developments for A5 and S5 were reproduced; these remain credited to him. F31's existing Lei-Zhang counterexample received a construction audit and remains credited prior work. The [literature ledger](#) and [scope report](#) retain these distinctions.

N3 produced an obstruction to a step in a published cover construction and to an attempted repair, together with a restricted argument whose countable case is prior. It did not settle the general question. F37 and F39 produced obstructions to algorithmic shortcuts rather than the requested algorithms. Exact F20 searches and equational attempts were inconclusive. These failures help explain the research process without enlarging the success count.

Implementation failures were also kept. A GAP zero exit status can coexist with an error, so recorded success required inspection of the output and its mathematical meaning.

Negative controls distinguished rational from integral splitting, matrices from braid equality and primitive leading pairs from complete integral branches. For B9 an early return comparison missed a matrix entry and was corrected in the historical record. The later uniform first-column obstruction is a different argument. Retaining both makes the correction inspectable.

## 8.2 What this experiment measures

The run produced a portfolio of arguments and implementations under an explicit wall-clock limit. It demonstrates how an agent can move between literature, calculations, proof construction and targeted formalization. It does not measure a probability of solving an arbitrary open problem. The input is historical, selection was adaptive, prior knowledge was available and status checks were necessarily incomplete. No blinded control or complete novelty survey was performed.

There are also different meanings of “complete”. F34(a) and F38(a) propose terminating algorithms while importing a full language construction that was not implemented. N5 and N8 have full candidate commands, but finite controls cannot establish their completeness proofs. N9 constructs one fixed ambient group effectively from an undecidable Diophantine polynomial without printing its entire numerical presentation. Each is a mathematical claim with a particular evidence boundary, not a uniform software milestone.

The next assessment should concentrate on the actual mathematical interfaces, including F38(c)’s shortening argument and N8’s full-block separation, and independently establish attribution. This version preserves the deadline record for that purpose. Later corrections should change the current assessment explicitly while leaving the historical counts and artifacts recoverable.

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## A The integral central-splitting decision

This supplies the abelian step of Proposition 6.1. Given a finitely generated abelian group  $Z$  and finitely many constraints  $v \in Z_i + dZ$ , decide whether  $Z = Z_1 \oplus Z_2$  satisfies them, including requirements that specified factors be nontrivial.

Write  $Z = F \oplus T$ , with  $F \cong \mathbb{Z}^n$  and  $T$  finite. Enumerate all direct splittings  $T = T_1 \oplus T_2$  and all homomorphisms  $\beta : F \rightarrow T$ . These are finite sets. Every ordered splitting of  $Z$  is the image of

$$Z_1 = A \oplus T_1, \quad Z_2 = B \oplus T_2, \quad F = A \oplus B$$

under a shear  $(f, t) \mapsto (f, t + \beta(f))$ . Indeed the intersections with  $T$  split  $T$ , the images in  $F$  split  $F$ , and chosen free summands are graphs of homomorphisms into the complementary torsion factor. Combining these graph maps gives  $\beta$ .

Apply the inverse shear to the constraints. Each separates into the finite test  $t \in T_i + dT$  and a free test  $f \in A_i + dF$ . Let  $W_i$  be the saturation of the lattice spanned by the exact ( $d = 0$ ) vectors assigned to side  $i$ . The exact conditions are possible precisely when

$$W_1 \cap W_2 = 0, \quad W_1 + W_2 \text{ is primitive in } F.$$

Test these by Smith form and choose  $F = W_1 \oplus W_2 \oplus C$ . Every splitting containing the  $W_i$  is obtained from one of

$$A_k = W_1 \oplus \langle c_1, \dots, c_k \rangle, \quad B_k = W_2 \oplus \langle c_{k+1}, \dots, c_e \rangle, \quad 0 \leq k \leq e,$$

where  $e = \text{rank } C$ , by an automorphism fixing  $W_1 + W_2$  pointwise. Such automorphisms have matrices

$$\Gamma = \left\{ \begin{pmatrix} I & 0 & U \\ 0 & I & V \\ 0 & 0 & W \end{pmatrix} : W \in \text{GL}_e(\mathbb{Z}) \right\}.$$

A finite generating set consists of the displayed shears, elementary matrices, signed permutations on  $C$ . Let  $M$  be the lcm of all positive moduli, or one if there are none. Enumerate the finite image of  $\Gamma$  modulo  $M$ , retaining an actual word in integral generators for each image. For each image and each  $k$ , compute the projection  $P$  onto the transported  $A_k$  along  $B_k$ . The remaining constraints are exactly

$$(1 - P)f = 0 \pmod{d} \quad \text{on side one}, \quad Pf = 0 \pmod{d} \quad \text{on side two}.$$

They are finite congruence tests. A passing stored word gives an actual integral splitting, not merely a modular idempotent; the free ranks and torsion sizes decide required nontriviality. Conversely every splitting is represented by one of these branches and residues. This proves the finite constructive step used in the group argument. In particular, idempotents with incompatible ranks at different primes cannot be mistaken for an integral splitting.

For the preceding Smith reduction, if  $UE_iV = D_i$ , apply  $U$  to the central defect vector. A zero row requires membership in  $Z_i$ ; a diagonal entry  $d > 0$  requires membership in  $Z_i + dZ$ . These operations are valid over every abelian group, including finite groups. Once a splitting passes, solving the same Smith system constructs corrected lifts and therefore the group factors.

The controlling [proof](#) and [integrality audit](#) include torsion shears, mixed centres, composite moduli and dependent-relator controls. The later [finite-presentation command](#) connects this stage to rational support and the complete quotient search.

## B The shortening interface for bounded equivalence

This appendix gives the substantive sufficiency step of Proposition 5.6. It is a candidate application of Sela's graded shortening theorem [29]; the JSJ and Rips machinery is imported, not verified by the finite stabilizer program.

### B.1 The relative lemma and its normalization

Let  $A = F(a_1, \dots, a_k)$ ,  $k \geq 2$ , and let  $u \neq 1$  lie in no proper free factor of  $A$ . The required assertion is that for fixed target rank  $r \geq 2$  there is  $K$  such that every injection  $f : A \rightarrow F_r$  admits  $c \in F_r$  and  $\theta \in \text{Aut}(A)$  fixing  $u$  with

$$\max_i |cf(\theta(a_i))c^{-1}| \leq K\|f(u)\|. \quad (12)$$

Adjoin a generator  $z$ , set  $G = A * \langle z \rangle$  and  $P = \langle u, z \rangle$ . There is no nontrivial free splitting with  $P$  in one factor. Indeed the intersection of that free factor with  $A$  is a free factor of  $A$  containing  $u$ , hence all of  $A$ ; it also contains  $z$ . The same intersection argument shows that every automorphism fixing  $P$  pointwise preserves  $A$  and restricts to  $\text{Aut}(A/u)$ .

Conjugate  $f(u)$  to a cyclically reduced word, then minimize  $\max_i |f(\theta(a_i))|$  over  $\text{Aut}(A/u)$ . The integer minimum exists. Failure of (12) would give minimizing injections  $f_m$  with

$$L_m = |f_m(u)| = \|f_m(u)\|, \quad M_m = \max_i |f_m(a_i)| > 2^m(1 + L_m).$$

Extend to injections  $h_m : G \rightarrow F_r * \langle z \rangle$  fixing  $z$ . Use the fixed generating set  $\{a_1, \dots, a_k, u, z\}$ , with the parameters explicit. The maps are shortest under the graded modular subgroup, because it fixes  $P$  and every such precomposition restricts to the minimization already made.

In Sela's Definitions 10.1–10.3, the proposed reduction regards this relative group as rigid or solid. If the graded JSJ is trivial it is rigid. Otherwise minimizing an embedding in its graded modular orbit gives a constant shortening sequence with quotient  $G$ , which dominates every quotient and is the maximal shortening quotient; constant sequences are allowed in the source's convention. The no-relative-free-product condition follows from injectivity and the free-factor argument above. The displayed  $2^m$  inequality then gives a flexible graded sequence. These are the hypotheses that must be checked in applying the deep proper-quotient conclusion, rather than replacing it by ordinary bounded-parameter shortening.

Here is the normalization check needed to identify its canonical limiting quotient. In the target tree let  $o$  be the identity,  $D_m(x)$  the maximum displacement by the chosen generators, and  $\mu_m = \min_x D_m(x)$  over vertices. The axes of  $f_m(u)$  and  $z$  meet precisely at  $o$ . One of them has projection of  $x$  equal to  $o$ , so  $D_m(x) \geq 2d(x, o)$ . A generator attaining  $M_m$  gives  $M_m \leq 2d(x, o) + D_m(x)$ . For a minimizing vertex  $x_m$ ,

$$M_m/2 \leq \mu_m \leq M_m, \quad d(x_m, o)/\mu_m \leq 1/2. \quad (13)$$

Moreover, for some  $s \in \{u, z\}$  along a subsequence,  $o \in [x_m, h_m(s)x_m]$ . Thus  $o$  and all its fixed orbit translates belong to finite orbit hulls in the minimum-displacement normalization; bounded distance alone is not being used to assume their survival. After rescaling by  $\mu_m$ ,  $P$  fixes the limiting point of  $o$ .

Pass to a subsequence with  $|f_m(a_i)| = M_m$  for one fixed  $i$  and  $M_m/\mu_m \rightarrow \rho \in [1, 2]$ . The three points  $h_m(a_i)o, h_m(za_i)o, h_m(z^{-1}a_i)o$  have unscaled pairwise distances  $2M_m + 1, 2M_m + 1, 2M_m + 2$ , by free-product normal forms. They give a nondegenerate tripod with all legs of length  $\rho$  in the same canonical limit. An element in the action kernel fixes this tripod. Sela's elementary tripod lemma places it eventually in the kernels of the  $h_m$  [29, Lemma 1.3(iii)]; injectivity makes it trivial. Hence the limiting quotient is  $G$  itself. The flexible inequality survives passage to a subsequence. This contradicts the properness of flexible graded quotients of rigid or solid groups [29, Lemma 10.4(ii)], giving (12).

Shortness here concerns the original maps with their fixed parameter images. Conjugating to the minimizing basepoint is used for the limit; it is not asserted to preserve based graded shortness. The separate [normalization supplement](#) records the complete comparison with the source convention.

## B.2 From a finite stabilizer orbit to the length bound

Suppose first that  $u$  fills  $A$  as a free factor and the stabilizer has a finite orbit on  $[v]$  in  $A$ , represented by words of lengths at most  $B$ . Applying (12) to an injection  $f$  and then to a representative of  $\theta^{-1}[v]$  gives

$$\|f(v)\| \leq BK\|f(u)\|.$$

No value of  $K$  is required by the eventual decision algorithm.

In ambient rank at least three choose a minimal free factor  $A$  containing  $u$ . If  $\text{rank } A \geq 2$ , every complementary letter  $b$  can be Nielsen-twisted as  $b \mapsto ba_1$  or  $b \mapsto ba_2$  for distinct basis letters  $a_1, a_2$  of  $A$ . Finiteness of  $H_u[v]$  forces bounded lengths under both iterations. For a cyclic word whose non- $e$  letters are  $t_1, \dots, t_s$  and intervening  $e$ -runs are  $e^{k_i}$ , the exact formula for  $d \mapsto de$  is

$$\|\sigma_{d,e}^n(w)\| = s + \sum_i |k_i + n(\mathbf{1}_{t_i=d} - \mathbf{1}_{t_{i+1}=d^{-1}})|. \quad (14)$$

Inverse non- $e$  neighbors cannot newly cancel: their intervening run was nonempty and its coefficient of  $n$  is zero. Thus zero growth requires each positive  $b$  to pair with a negative  $b$  across a pure  $a_j$  run. It cannot satisfy this for both  $a_1, a_2$  unless no  $b$  occurs. Hence  $v$  is conjugate into  $A$ , and the preceding bound applies to each restricted ambient automorphism.

If  $A = \langle a \rangle$  and the ambient rank is at least three, use the maps  $b \mapsto bc$ ,  $c \mapsto cb$ ,  $b \mapsto ba$ , and  $x_j \mapsto x_j b$  for the remaining basis letters. Formula (14) shows that their common bounded conjugacy classes are exactly powers of  $a$ . One explicit check is the product of the first three two-vertex zero-growth graphs: its  $b, c$  edges form a forest, leaving only  $a$ -loops and loops of the unused generators after pruning. The last maps remove those unused letters. The full eight-vertex graph is listed in the retained [Nielsen argument](#). Therefore  $v$  is a power of the same primitive element, with constant length ratio.

Finally take ambient rank two and  $u = a^p$ . Bounded iterates under  $b \mapsto ba$  put  $v$ , up to conjugacy, in  $\langle a, bab^{-1} \rangle$ . Fix an expression of length  $L$  in these generators. Under an automorphism put  $A = \alpha(a)$ ,  $B = \alpha(b)$ ,  $C = BAB^{-1}$  and  $\ell = \|A\| \geq 1$ . The rank-two commutator identity gives  $\|AC^{-1}\| = \|ABA^{-1}B^{-1}\| = 4$ . If the axes of  $A, C$  are disjoint their distance is at most one, by the tree product formula. At a nearest vertex their displacements are at most  $\ell + 2$ . Therefore

$$\|\alpha(v)\| \leq L(\ell + 2) \leq 3L\ell = \frac{3L}{|p|} \|\alpha(u)\|.$$

This is specifically an ambient-automorphism argument. It completes the one-sided criterion. Applying it in both directions gives commensurability of the two oriented stabilizers.

## C The all-class commutator recursion

This appendix develops the candidate proof of Proposition 6.2. The untruncated rational free Lie algebra is  $L = L_{\mathbb{Z}} \otimes \mathbb{Q}$ , graded by weight. Hall group coordinates remain integral throughout; rational changes below prove structural separation and do not replace the integral solution lattice. The elementary cases  $g = 1$ , rank at most one and class at most one are handled directly.

### C.1 Complete leading branches

The exact moves  $(x, y) \mapsto (yx, y)$  and  $(x, y) \mapsto (x, xy)$  preserve  $[x, y]$ . The rotation  $(x, y) \mapsto (y^{-1}, y^{-1}xy)$  does so as well. On a common leading layer these realize  $\mathrm{SL}_2(\mathbb{Z})$ . If the two leading terms are dependent, Euclidean reduction kills one of them and increases its weight. Nonzero commuting elements of an untruncated free Lie algebra are proportional: otherwise they span a two-dimensional abelian subalgebra, contrary to Shirshov–Witt. The normalized nonidentity target therefore has leading branches

$$[C, D] = W, \quad C \in (L_{\mathbb{Z}})_p, \quad D \in (L_{\mathbb{Z}})_q, \quad p \leq q, \quad p + q = d.$$

For  $p < q$  the bracket has no zero divisors, also over an algebraic closure. Its projective linear projection on  $\mathbf{P}(L_p) \times \mathbf{P}(L_q)$  has centre disjoint from that Segre variety. A positive-dimensional projective fibre would meet the centre hyperplane in its ambient fibre space, so fibres are finite. Normalize a first nonzero coordinate of  $C$  to one in finitely many charts. Gröbner bases give zero-dimensional coordinate algebras; rational roots of coordinate-multiplication characteristic polynomials, followed by testing the equations, enumerate all rational points. For each first-factor line choose primitive integral  $C_0$ . The solution  $D_0$  of  $[C_0, D_0] = W$  is unique. Integral factors exist only if  $D_0$  is integral, and are then exactly

$$(kC_0, D_0/k), \quad k \text{ a nonzero signed divisor of the content of } D_0.$$

For  $p = q$ , the bracket map  $\Lambda^2 L_p \rightarrow L_{2p}$  is injective: split associative words into two length- $p$  blocks to recover  $C \otimes D - D \otimes C$ . The exterior preimage of  $W$  must be integral and have alternating rank two. Intersect its rational support with  $L_{\mathbb{Z}}$  to get a saturated rank-two lattice. In an oriented basis the exterior tensor is  $he_1 \wedge e_2$ ,  $h > 0$ . Enumerate the finitely many index- $h$  sublattices by Hermite form. Every other oriented basis is reached by the exact pair moves, leaving higher coordinates unrestricted. This proves finite completeness of the leading list, not just the existence of some leading decomposition.

## C.2 Correction kernels and exact periods

Fix  $C, D$ , and put  $n = c - d$ . At offset  $s$  the integral correction map is  $A_s(U, V) = [U, D] + [C, V]$  on the full components of weights  $p + s, q + s$ . We use the two-coefficient inner-solution theorem: if  $a, b$  belong to a free basis and  $[a, V] + [U, b] = 0$ , then  $U, V \in \text{Lie}(a, b)$  [26, 1]. It is not applied to arbitrary coefficients without first constructing a free basis containing them.

For  $0 < s < q - p$ , a kernel has dimension at most one. For nonzero  $U$ , the graded subalgebra

$$\mathbb{Q}C + \mathbb{Q}U + \bigoplus_{j \geq p+s+1} L_j$$

has a homogeneous free basis containing  $C, U$ , and contains  $D, V$ . The inner-solution theorem puts  $D, V$  in  $\text{Lie}(C, U)$ . Two independent first components  $U_1, U_2$  would similarly give  $D \in \text{Lie}(C, U_1) \cap \text{Lie}(C, U_2) = \mathbb{Q}C$ , impossible by weight. These nonzero one-dimensional kernels are called exceptional. At  $s = q - p > 0$  the kernel is  $\mathbb{Q}(D, 0)$ : use  $\mathbb{Q}C + \bigoplus_{j \geq q} L_j$ , kill  $C$  and higher generators, and apply the centralizer fact. Its period is the exact substitution  $x \mapsto y^k x$ , with all residues retained if  $D$  is nonprimitive. After that offset every kernel pair lies in  $\text{Lie}(C, D)$ , by applying the same theorem in  $\mathbb{Q}C + \mathbb{Q}D + \bigoplus_{j \geq q+1} L_j$ ; for equal weights use  $L_{\geq p}$ .

Here is a finite period construction for all these latter kernels. In the free Lie algebra on  $a, b$  with weights  $p, q$ , truncated at  $c$ , set  $W_0 = \log(e^{-a}e^{-b}e^ae^b)$ . There is a filtered automorphism  $\sigma$ , with identity associated graded, satisfying  $\sigma([a, b]) = W_0$ . Successively remove homogeneous discrepancies by correcting  $a, b$ : every derived Lie element is in  $[a, H] + [H, b]$ , by Jacobi induction. This is the degree-wise symplectic-expansion construction; the related prior construction is credited in [19] and the retained gauge audit.

If  $[u, b] + [a, v] = 0$  at offset  $s$ , the positive-weight derivation  $\delta(a) = u$ ,  $\delta(b) = v$  kills  $[a, b]$ . Thus

$$\alpha = \sigma \exp(\delta) \sigma^{-1}$$

fixes  $W_0$  and has first increments  $u, v$ . Its powers have rational polynomial coefficients in the exponent. In the weighted two-generator Hall group obtained by discarding weights greater than  $c$ , convert  $\exp(\alpha^k(a)), \exp(\alpha^k(b))$  to group coordinates. Choose  $M > 0$  divisible by all denominators of their nonconstant coefficients. At  $k = \pm M$  the coordinates are integral, so  $\alpha^M$  and its inverse give actual group automorphisms fixing  $[A, B]$ . Substitution  $A \mapsto x$ ,  $B \mapsto y$  gives reversible pair substitutions in the target group, not necessarily automorphisms of that target.

Noncommuting  $C, D$  freely generate their subalgebra. Express a complete rational kernel basis in them and apply this construction to each integer kernel direction. The resulting integral multiples span a full-rank exact period lattice. Each such layer therefore has finitely many residue branches. Replacing the entire lattice by one rational representative would not be valid.

### C.3 A diagonal positional-polynomial lemma

Let  $E_0, E_1, \dots$  freely generate an associative algebra over  $\mathbb{Q}$ , with  $\delta(E_i) = E_{i+1}$ . Encode the coefficient of  $E_{i_1} \cdots E_{i_m}$  as that of  $z_1^{i_1} \cdots z_m^{i_m}$ . This is a linear dictionary, not an algebra homomorphism;  $\delta$  becomes multiplication by  $z_1 + \cdots + z_m$ .

Suppose  $\delta V = [E_0, D]$  and  $P$  encodes an  $m$ -letter  $D$ . Then  $V$  has encoding

$$\frac{P(z_2, \dots, z_{m+1}) - P(z_1, \dots, z_m)}{z_1 + \cdots + z_{m+1}}. \quad (15)$$

Polynomiality implies cyclic invariance of  $P(y_1, \dots, y_m)$  on the hyperplane  $y_0 + \cdots + y_m = 0$ . Let  $R$  denote restriction of an  $(m+2)$ -letter encoding to

$$z_{m+2} = z_1, \quad z_1 + \cdots + z_{m+2} = 0.$$

**Lemma C.1.** *If  $R([E_0, V]) = 0$ , then  $P$  is constant.*

*Proof.* Put  $z = (z_1, \dots, z_m)$  and  $a = -\frac{1}{2} \sum z_i$ . Substituting (15) into the assumption gives

$$2P(z) = P(z_2, \dots, z_m, a) + P(a, z_1, \dots, z_{m-1}).$$

The calculation may divide by  $a$  on the nonempty open set  $a \neq 0$ , then extends polynomially. Cyclic invariance rewrites it as

$$2P(z) = P(z + ae_1) + P(z + ae_m).$$

On the zero-sum hyperplane, let  $T_{i \rightarrow j}$  halve coordinate  $i$  and add the other half to  $j$ . The last identity says  $P$  is the average of its values under the two transfers from coordinate zero to its cyclic neighbors. Cyclic invariance gives the same identity at each coordinate. All these maps preserve the hyperplane and do not increase the  $\ell^1$  norm.

On a closed  $\ell^1$  ball,  $P$  attains a maximum. At a maximizing point both terms of each average must again maximize, so every finite transfer composition preserves maximality. Repeat a complete clockwise cycle of transfers  $m+1$  times. Its matrix  $B$  is positive and column-stochastic: retained halves permit waiting, and clockwise paths carry positive mass between every pair of coordinates. If  $\epsilon$  is its smallest entry and  $\ell = m+1$ , write  $B = \epsilon J + (1 - \ell\epsilon)C$  with  $C$  column-stochastic when the latter coefficient is positive. On the zero-sum hyperplane  $Jy = 0$ , so  $B$  strictly contracts  $\ell^1$ ; if that coefficient is zero it annihilates the hyperplane. Iteration tends to zero, making the maximum  $P(0)$ . The same applies to the minimum. Every ball therefore has constant value.  $\square$

For a Lie element  $D$ , a nonzero constant positional polynomial would be a multiple of  $E_0^m$ , which is a Lie element only for  $m = 1$ . Except for  $D = \lambda E_0, V = 0$ , the nonzero Lie solution therefore has  $R([E_0, V]) \neq 0$ . The other two identities needed below are

$$R(\delta F) = 0, \quad R([E_j, F]) = z_1^j R([E_0, F]). \quad (16)$$

They follow immediately from the restriction and apply at each fixed letter count. Injectivity of  $\delta$  on positive-length tensors is injectivity of multiplication by their positional sum.

### C.4 Why all later columns fail to cancel the first quadratic

Fix all pair coordinates below the first exceptional offset  $t < q - p$ . If  $n \geq 2t$ , keep all coordinates at offsets  $t, \dots, 2t - 1$  and solve their commutator equations jointly. They are an affine integer linear system: products of two increments first enter at offset  $2t$ . Its complete solution set is empty or an affine integer lattice  $\mathcal{S}$ . The first kernel is a line. If its coefficient

varies on  $\mathcal{S}$ , choose an integral lattice basis with that coefficient  $k_0 + bT$ ,  $b \neq 0$ , and every other direction starting strictly later. At offset  $2t$ , the new error has the form

$$q(T) + B\mathbf{s} = 0 \pmod{\text{im } A_{2t}},$$

with  $q$  of degree at most two and  $B$  constant. The needed statement is

$$q_2 \notin \text{span}_{\mathbb{Q}}(\text{im } A_{2t}, \text{ all columns of } B). \quad (17)$$

Showing merely  $q_2 \notin \text{im } A_{2t}$  would leave an essential gap.

Write  $X = \log x$ ,  $Y = \log y$ . In the free graded subalgebra  $L_{\geq p}$  include  $C$  in a homogeneous free basis. A fixed filtered substitution, identity on associated graded, removes the fixed  $X$ -components at offsets  $1, \dots, t-1$ . This is a rational proof normalization independent of the parameters. Put

$$Z = h(\text{ad}_X)Y, \quad h(z) = (1 - e^{-z})/z.$$

The part of  $\log[x, y]$  linear in  $Y$  is  $[X, Z]$ . Every other term has at least two  $Y$  occurrences and one  $X$ ; any dependence on a block parameter has weight at least  $p + 2q + t > d + 2t$ . Such terms may contribute to the fixed target but do not affect its varying block. The required components of  $X, Y, Z$  are affine in the block parameters: two varying  $X$  increments already have weight  $2(p + t) > p + 2t$ , and the analogous inequalities handle  $Y$  and  $h(\text{ad}_X)Y$ .

Use the graded free subalgebra  $\mathcal{A} = \mathbb{Q}C + \bigoplus_{j \geq p+t} L_j$ . Include  $C$  and the nonzero first kernel component  $U$  in a homogeneous free basis and project onto  $\text{Lie}(C, U)$ . The inner-solution theorem puts the leading  $D$  and corresponding  $V$  in this subalgebra, so the projection fixes them. Including all of  $L_{p+t}$  also retains arbitrary particular solutions at that offset; it is not a projection justified only for the kernel line.

Set  $E_j = (\text{ad}_C)^j U$ ,  $\delta = \text{ad}_C$ . Lazard elimination makes the ideal of  $U$  free on these  $E_j$ . Let  $D_m$  be the nonzero component of  $D$  with maximal  $E$ -letter count. Its associated  $V_{m+1}$  satisfies  $\delta V_{m+1} = -[E_0, D_m]$ . The harmless sign change in Lemma C.1 yields

$$R([E_0, V_{m+1}]) \neq 0.$$

The constant exception is excluded by the Lie condition if  $m > 1$  and by  $q > p + t$  if  $m = 1$ .

Every projected first-factor component at offset  $i \leq 2t$  has at most one  $E$  letter, since  $2(p + t) > p + 2t$ . Thus it is zero or a multiple of  $E_j$  with  $pj = i - t$ . Consequently  $[L_{p+2t}, D]$  has at most  $m + 1$  such letters, and  $R$  kills  $[C, L_{q+2t}]$ . At letter count  $m + 2$  it annihilates  $\text{im } A_{2t}$ .

For  $0 \leq j < t$ , let  $F_j$  be the letter-count- $(m + 1)$  part of the fixed  $Z$  component at offset  $j$ ;  $F_0 = 0$ . The coefficient of  $T$  in the already satisfied equation at degree  $d + t + j$ , followed by projection and  $R$ , gives the triangular convolution

$$bR([E_0, F_j]) + \sum_{i=t+1}^{t+j} b_i z_1^{(i-t)/p} R([E_0, F_{t+j-i}]) = 0.$$

Only integral exponents occur. Induction gives  $R([E_0, F_j]) = 0$  for all  $j < t$ . At degree  $d + 2t$ , any other parameter starts later than  $t$ : its varying  $X$  part pairs only with such fixed  $Z$  parts, and its varying  $Z$  part pairs with  $C$  or a vanished fixed lower  $X$  part. Equation (16) therefore kills every column of  $B$ , including universal-kernel columns.

The coefficient of  $T^2$  is  $b^2[U, V]$ , whose letter-count- $(m + 2)$  part has nonzero  $R$ -image. This proves (17). After the lower equations have been satisfied, the target error begins in degree  $d + 2t$ . Hall and logarithmic coordinates agree there, and the filtered normalization is identity in that first error degree. Thus the functional pulls back to the actual group correction space. The retained [coefficient dictionary](#) spells out this group interface and the later columns.

## C.5 Termination and completeness

For each leading pair process successive offsets. A zero kernel has at most one integral affine solution. A universal kernel has finitely many branches modulo its full-rank exact period lattice. At the first exceptional offset  $t$ , if fewer than  $2t$  offsets remain from the leading degree, the whole remaining system is jointly linear and is decided at once.

Otherwise form  $\mathcal{S}$ . Reject an empty lattice. If its first exceptional coefficient is constant, fix the prefix through  $t$  and restart at  $t + 1$ . If it varies, use rational linear algebra to find a functional annihilating in  $A_{2t}$  and all later columns but not  $q_2$ , as guaranteed by (17). Its scalar quadratic has at most two integral roots. For each root fix the entire prefix through  $t$  and restart, freeing higher coordinates again.

Freeing those coordinates may repeat unsuccessful continuations, but cannot lose a solution. A positive answer still requires every equation through class  $c$  and a final actual commutator check. The nonunit step  $b$ , all integer congruences and all exact-period residues remain in the corresponding integral systems. A rational correction or rational root is never accepted as an integral one. Each restart increases the fixed offset and every branch has finitely many children, so the tree terminates. Conversely any solution has a normalized leading branch and survives every necessary equation, exact period and quadratic root choice. This proves the proposed decision and construction.

The proof needs no bound on the number of exceptional offsets and no integer-curve oracle. The positional argument has substantial Lean support, including its ordered-word dictionary and full correction-span formulation. The abstract free-Lie identification, the production of the group block and the full recursive program are outside Lean. Higher-weight positive and negative native group controls support specific instances; they do not establish the structural proof in all ranks and classes.

## D Certificates, implementations and formalization limits

### D.1 The final growth certificate

A bridge atom has no internal level at which just one vertical edge has nonzero flow. Such an edge must have coefficient  $+1$  by total flux. In a product of atoms, exactly these separating cuts mark the factor boundaries. Restriction to successive edge bands recovers their flows and endpoints. Thus distinct atoms of arbitrary positive heights freely generate a monoid. For a specified alphabet with cost series  $F(t)$ , its positive root  $F(R) = 1$  gives  $\lambda_2 \geq 1/R$ ; for infinite alphabets exhaust by finite subsets. Costs are lengths of supplied representatives, not asserted geodesic lengths.

The initial certificate has 21,483 distinct atoms of costs at most 14, with count vector

$$(1, 2, 2, 2, 2, 6, 26, 71, 162, 341, 779, 1961, 5098, 13030).$$

Write  $a$  for the vertical and  $b$  for the horizontal generator. For height at least two, the operations  $P_{s,t}(w) = ab^s a^{-1} w b^t$  preserve atom status. The fixed initial vertical edge cancels as indicated in a bridge representative. Removing the horizontal boundary lines at heights one and  $H$ , and normalizing the remaining flow by its first internal vertical support, classifies the resulting  $\mathbb{Z}^2$  orbits. Equality of normalized cores plus endpoint coordinates implies equality of full flows: their difference would be a finitely supported circulation on two disjoint lines, hence zero. Height one is the separate family  $ab^m$ .

For a retained orbit with seed coordinates  $(x_j, z_j)$  and costs  $c_j$ , the cost at  $(x, z) \in \mathbb{Z}^2$  is

$$f(x, z) = \min_j (c_j + |x - x_j| + |z - z_j|).$$

Outside the seed bounding rectangle, clamping either coordinate adds exactly its outside distance. The central rectangle, four strips and four quadrants therefore give a rational series  $F_{\text{old}}(t) = C(t) + S(t)/(1-t) + Q(t)/(1-t)^2$ , including the height-one series. Shortened representatives add 320 seeds to the original list, with the same full flows and verified smaller costs. The finite contracted-grid support-connection calculation uses the credited flow/Euler construction [24]; its polynomial data are retained in the [completion manifest](#).

For height  $H \geq 3$ , add the Nielsen shear  $\alpha_k(a) = ab^k$ ,  $\alpha_k(b) = b$ . It commutes with both boundary operations. If  $x_1, x_2$  are the minimum horizontal coordinates of nonzero vertical support at internal levels one and two, and  $(H, y)$  is the endpoint, put

$$d = x_2 - x_1, \quad u = x_1 - d, \quad v = y - x_1 - (H - 1)d.$$

The boundary action changes  $(u, v)$  freely, while  $\alpha_k$  changes  $d$  to  $d + k$ . Applying  $\alpha_{-d}$  and  $P_{-u, -v}$  and retaining the *full* normalized flow classifies the free  $\mathbb{Z}^3$  action. In each such orbit the old alphabet occupies finitely many complete  $d$ -planes. Add one tail strictly beyond each extremal plane, with all boundary coordinates. The two tails and all old planes are disjoint by their coordinates; different normalized flows give disjoint orbits.

A chosen seed with  $V$  vertical letters and initial tail cost  $L$  contributes

$$\frac{t^L}{1 - t^V} \left( \frac{1 + t}{1 - t} \right)^2.$$

The 21,803 retained seeds give 1,453 shear orbits and 2,906 tails. Their aggregated additional series is

$$F(t) - F_{\text{old}}(t) = \left( \frac{1 + t}{1 - t} \right)^2 \left( \frac{80t^{19} + 236t^{20} + 1658t^{21}}{1 - t^7} + \frac{68t^{22} + 448t^{23}}{1 - t^9} + \frac{6t^{22} + 48t^{23} + 362t^{24}}{1 - t^{10}} \right).$$

Exact arithmetic verifies

$$F(10^9/2676891785) \geq 1, \quad F(10^9/2676891786) < 1.$$

Only the lower endpoint of the resulting reciprocal-root interval is a lower bound on group growth; its upper endpoint is not a group upper bound. The independent upper certificate lists 648 words replaceable by shorter or lexicographically smaller equal-flow words. Every shortlex least representative avoids them. Its 1,968-state avoidance automaton has adjacency matrix  $A$  and an explicit positive integer vector  $v$  satisfying

$$Av \leq \frac{2943737759}{10^9} v.$$

This bounds accepted words, hence group balls, at that exponential rate. Separate GAP checks reconstruct all relations, transitions and integer inequalities. The [shear supplement](#) and [upper-bound audit](#) link the exact certificates and commands.

## D.2 Selected verification scopes

| Entry   | Retained evidence                                                                                                                      | Limit of that evidence                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| F28     | Exact matrix identities, GAP subgroup indices and a universal Lean integral recurrence.                                                | Free-group faithfulness and the invariant-subgroup interpretation remain written.           |
| M0      | Explicit primitive words, bases and inverse bases over six finite fields; separate GAP evaluation; terminating witness construction.   | Finite examples do not prove the all-rank implication; effective bounds need not be usable. |
| H4      | Exact finite torsion-class counts and native free-kernel controls.                                                                     | The unbounded output lower bound is the written argument.                                   |
| F34/F38 | 1,433 equation tuples, 18,072 literal and 2,300 native equation checks; polynomial-span and supplied-grammar checks.                   | The full constrained equation-to-grammar generator is unimplemented.                        |
| F38(c)  | Whitehead stabilizers and finite mod-three orbit procedure.                                                                            | No computational validation of graded shortening or its application.                        |
| N5      | Finite-group comparisons, mixed central constraints and the full promised-presentation command; actual returned factor words.          | Rational-support uniqueness and finite completeness are mathematical dependencies.          |
| N8      | Full word-input solver; integral leading branches, exact period residues, native group columns and positive/negative quadratic blocks. | The complete all-rank structural proof is not independently verified.                       |
| N9      | Integer-kernel saturation, actual GAP retractions for a small circuit, isolated-input checks and Lean normalization.                   | Neither DPRM nor the full fixed-group reduction is formalized here.                         |
| G9      | All retained flows and series, 26,154 tail/boundary controls, 2,906 shear compositions and exact rational inequalities.                | Infinite distinctness and the growth argument are written; the exact constant is unknown.   |
| B9      | Shelf identities, faithful Artin-action controls and separating Burau entries; complete partner decisions for 52 first colors.         | Burau is used to distinguish, not certify equality; no exhaustion of $B_4$ .                |
| F41/GA3 | Counting and separator controls, plus source-specific dependency audits.                                                               | No finite check proves the imported definable-set or tree-collapse theorems.                |

The N8 higher-weight negative control is illustrative. In rank two and class eleven, the full solver rejects all four normalized weight types; the two surviving leading branches give quadratics  $T^2 - 2T + 2$  and  $T^2 + 2T + 2$ . Native GAP reconstructs both integer kernels, 64 actual group columns and the two polynomials. It does not independently recompute the projective leading-pair lists. The complete [negative audit](#) states that boundary explicitly.

The formal environment was Lean 4.24.0 with pinned Mathlib. The [guide](#) gives exact revisions. An axiom audit and `--trust=0` do not mean every cached Mathlib artifact was rebuilt or checked by a second kernel. Formalization claims in this paper are confined to

their stated fragments.

## E Protocol, provenance and reproduction

### E.1 Authorization and dates

The organizers’ launch instruction, preserved in [the session record](#), was:

great, now proceed as in the kourovka run: Solve as many previously unsolved problems as you can (from the dataset you just created) within an overall time limit of 48 hours. The gap system is again installed here but you can also download and install further SW if useful. In general, try to limit yourself to 20 CPU cores and 100G RAM on this machine while working. Write detailed plans and reports, iterate fast and use the local git repo keeping your work and documenting your progress, commit often there (but don’t try to push). Work rigorously - your solutions will be independently reviewed.

Line wrapping and the doubled space before “within” are normalized here; the session JSON preserves the exact text. The launch record names GPT-6 Astra with xhigh reasoning as supplied by the user; the runtime model identifier was not independently verified. No subagents or outside mathematical referee participated in the recorded run.

| Record                 | UTC time or revision                                     |
|------------------------|----------------------------------------------------------|
| Start                  | 2026-09-28 10:04:49.670358                               |
| Research cutoff        | 2026-09-30 10:04:49.670358                               |
| Last research commit   | 401213c, 2026-09-30 09:47:18                             |
| Freeze observation     | 2026-09-30 10:05:47.903924                               |
| Administrative handoff | 0d6b855                                                  |
| Paper                  | Version 1, 30 September 2026; post-experiment exposition |

The final scope rendering preserves all 195 catalogue entries, all counts and all 96 bound inputs from the pre-deadline assessment. Its [snapshot](#) retains exact source bindings and the actual process observation. The paper does not rerun mathematical solvers or silently strengthen that assessment. Its build and editorial checks have their own records under [paper/data/](#).

### E.2 Where to start in the repository

The repository is <https://github.com/JUrban/gworld1> [7]. Unlike the filtered public Kourovka history, the cited GroupWorld research and handoff commit identifiers are directly available there. The root README links this PDF, its source, the original collection and the research reports.

| Record                             | Purpose                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------|
| <a href="#">Final report</a>       | Deadline outcome, prior coverage, failed work and integrity limits.                   |
| <a href="#">Review guide</a>       | Controlling proofs, supplements, source audits and their exact scopes.                |
| <a href="#">Scope ledger</a>       | Whole entries, named parts, credited prior answers and all uncounted IDs.             |
| <a href="#">Reproduction guide</a> | Pinned dependencies, selected commands, historical source recovery and omitted files. |

| Record                            | Purpose                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------|
| <a href="#">Catalogue</a>         | Original preparation index; its initial status labels are not the final assessment. |
| <a href="#">Source manifest</a>   | Archived problem pages, retrieval records and hashes.                               |
| <a href="#">Literature ledger</a> | Dated primary-source comparisons and reading limits.                                |
| <a href="#">Research log</a>      | Intermediate arguments, corrections and process chronology.                         |

The full argument and audit of an entry live under `problems/ID/`. Exact source and result versions are bound by manifests under `research/certificates/` and `research/audits/`; terminal command receipts are in `results/`. Historical hash mismatches must be resolved through the retained exact-version recovery indices, rather than treating an old manifest as certification of the current working file.

### E.3 Software and retained data

Recorded computation used Python 3.12.3, GAP 4.16.1, SymPy 1.14.0, mpmath 1.3.0 and python-flint 0.9.0. GAP package versions are in the guide. A fresh replay should use a separate checkout, new output names and separately dated review records. The original research runner refuses jobs after the deadline; resetting its clock would falsify the experiment. Direct review commands in the guide are distinct from the historical bounded research invocations.

At the final handoff all 6,395 reachable Git blobs were below 90,000,000 bytes; the largest was 75,046,692 bytes. This does not mean the software installation and every generated file are bundled. The 91 intentionally untracked bound paths are 88 A5 dependency sources and three B9 compiled binaries, with source/build records. A 127,724,170-byte formatted F34/F38 JSON was omitted; the retained 19,478,346-byte compact JSON has the same parsed data, and the precise formatting and hash permit recovery of the former bytes. These omissions are disclosed even though no history filtering was needed for this GroupWorld publication.

The paper itself needs only a LaTeX installation. Its build script records the TeX invocation and source/PDF hashes. The source archive includes the bibliography output and all included TeX files; it does not require GAP, Lean, the source problem website or a research certificate to compile. Mathematical reproduction is a separate task.

### E.4 Authorship and acknowledgments

Michael Kinyon and Josef Urban organized the experimental format. Vladimir Shpilrain hosts the collection and suggested this follow-up. The AI author denotes the system that generated the research arguments, programs and this manuscript under the organizers' instructions; it does not denote an independently accountable mathematical referee. The author list does not assert that every human author has verified every proof. The research-visit and funding acknowledgments appear with Kinyon's and Urban's names on the first page, following the preceding paper's format.